American International University- Bangladesh (AIUB)
Faculty of Engineering (EEE)

Course Name:	Microprocessor and Embedded Systems	**Course Code:**	COE3104
Semester:	Summer 2024-2025	**Sec:**	K
Lab Instructor:	Md. Ali Noor		

Experiment No:	03
Experiment Name:	Timers: Implementation of A Traffic Control System

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Performance Date:	23-7-25	**Due Date:**	08-08-25

Marking Rubrics (to be filled by Lab Instructor)

Category	Proficient [6]	Good [4]	Acceptable [2]	Unacceptable [1]	Secured Marks
Theoretical Background, Methods & procedures sections	All information, measures and variables are provided and explained.	All Information provided that is sufficient, but more explanation is needed.	Most information correct, but some information may be missing or inaccurate.	Much information missing and/or inaccurate.	
Results	All of the criteria are met; results are described clearly and accurately;	Most criteria are met, but there may be some lack of clarity and/or incorrect information.	Experimental results don't match exactly with the theoretical values and/or analysis is unclear.	Experimental results are missing or incorrect;	
Discussion	Demonstrates thorough and sophisticated understanding. Conclusions drawn are appropriate for analyses;	Hypotheses are clearly stated, but some concluding statements not supported by data or data not well integrated.	Some hypotheses missing or misstated; conclusions not supported by data.	Conclusions don't match hypotheses, not supported by data; no integration of data from different sources.	
General formatting	Title page, placement of figures and figure captions, and other formatting issues all correct.	Minor errors in formatting.	Major errors and/or missing information.	Not proper style in text.	
Writing & organization	Writing is strong and easy to understand; ideas are fully elaborated and connected; effective transitions between sentences; no typographic, spelling, or grammatical errors.	Writing is clear and easy to understand; ideas are connected; effective transitions between sentences; minor typographic, spelling, or grammatical errors.	Most of the required criteria are met, but some lack of clarity, typographic, spelling, or grammatical errors are present.	Very unclear, many errors.	
Comments:				Total Marks (Out of):	

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Title: **Timers: Implementation of A Traffic Control System**

Introduction:

Traffic control systems are vital components of modern urban infrastructure, ensuring the orderly and efficient movement of vehicles and pedestrians. The increasing complexity and volume of traffic in cities necessitate the use of advanced technologies for effective traffic management. One such technology involves the use of microprocessors and timers to automate and optimize traffic light operations.

In this lab experiment, the implementation of a traffic control system using an Arduino Mega, a popular microcontroller platform known for its versatility and ease of use has been explored. The Arduino Mega is equipped with a variety of features, including multiple timers, which can be harnessed to create precise and reliable timing sequences essential for traffic light control.

So, it can be said that the main objectives of this experiment are to:

1. Get familiarized with the implementations of Arduino UNO/Mega microcontroller.
2. Get familiarized with Timer.
3. Implement the functionality of Timer0 through programming.
4. Implement a basic Traffic Light Control System.

Theory and Methodology:

Timer: Every electronic component of a sequential logic circuit works on a time base. This time base helps to keep all the work synchronized. Without a time base, devices would have no idea as to when to perform a particular action. Thus, the timer is an important concept in the field of electronics.

A timer/counter is a piece of hardware built into the Arduino controller. It is like a clock and can be used to measure time events. A timer is a register whose value increases/decreases automatically.

In AVR, timers are of two types: 8-bit and 16-bit timers. In an 8-bit timer, the register used is 8-bit wide whereas, in a 16-bit timer, the register width is 16-bits. This means that the 8-bit timer is capable of counting $2^8 = 256$ steps starting from 0 to 255. Similarly, a 16-bit timer is capable of counting $2^{16} = 65536$ steps starting from 0 to 65535.

Apparatus:

1. Arduino Uno / Arduino Mega [1]
2. LED Lights (YELLOW [1], RED [1], GREEN [1])
3. Resistors (220 ohms) [3]
4. Breadboard [1]
5. Jumper Wires
6. Proteus Software
7. Arduino IDE

Experimental Setup:

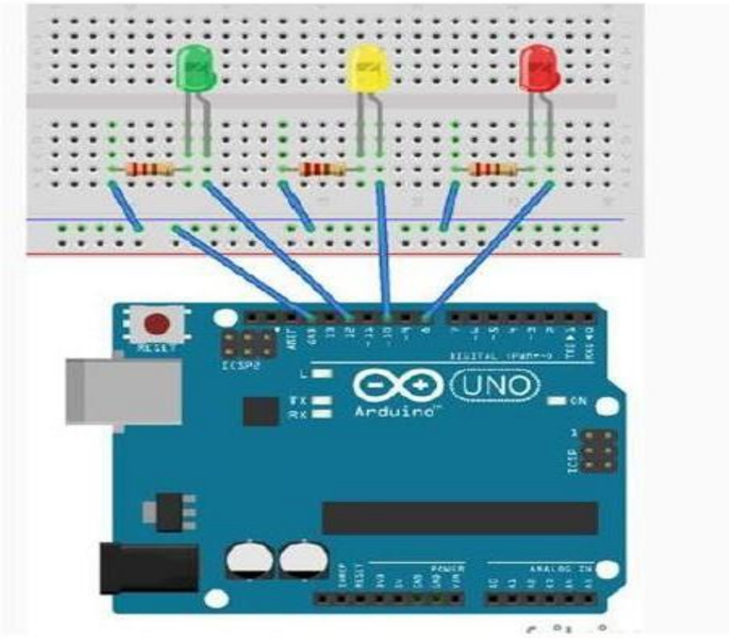


Figure 1: Experimental Setup a TLC using Arduino Mega (Timer0)

Hardware Implementation:

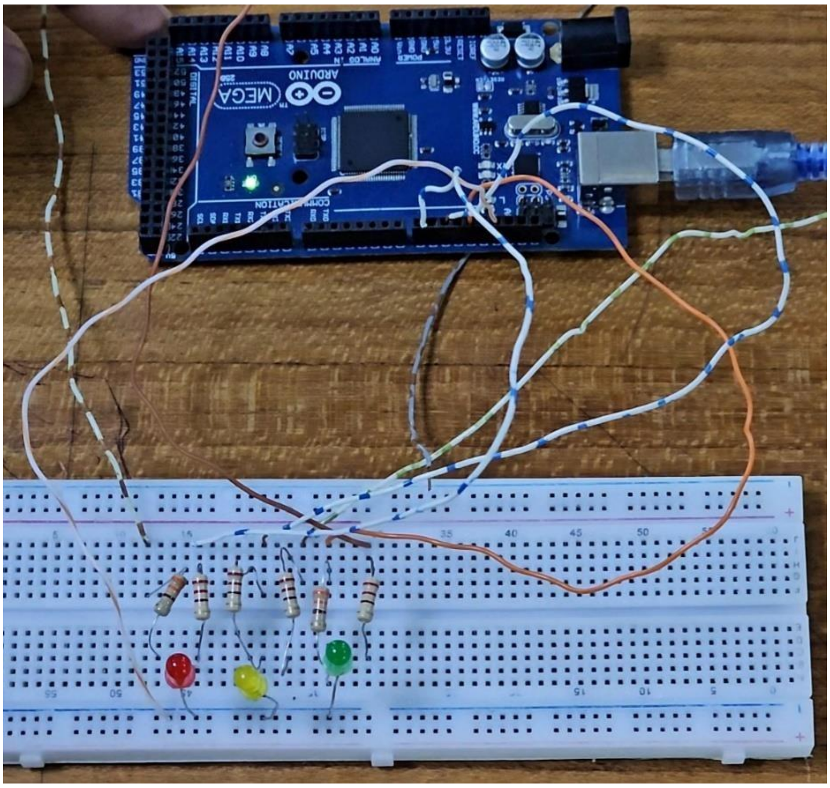


Figure 2: Hardware Implementation of a TLC using Arduino Mega

Simulation:

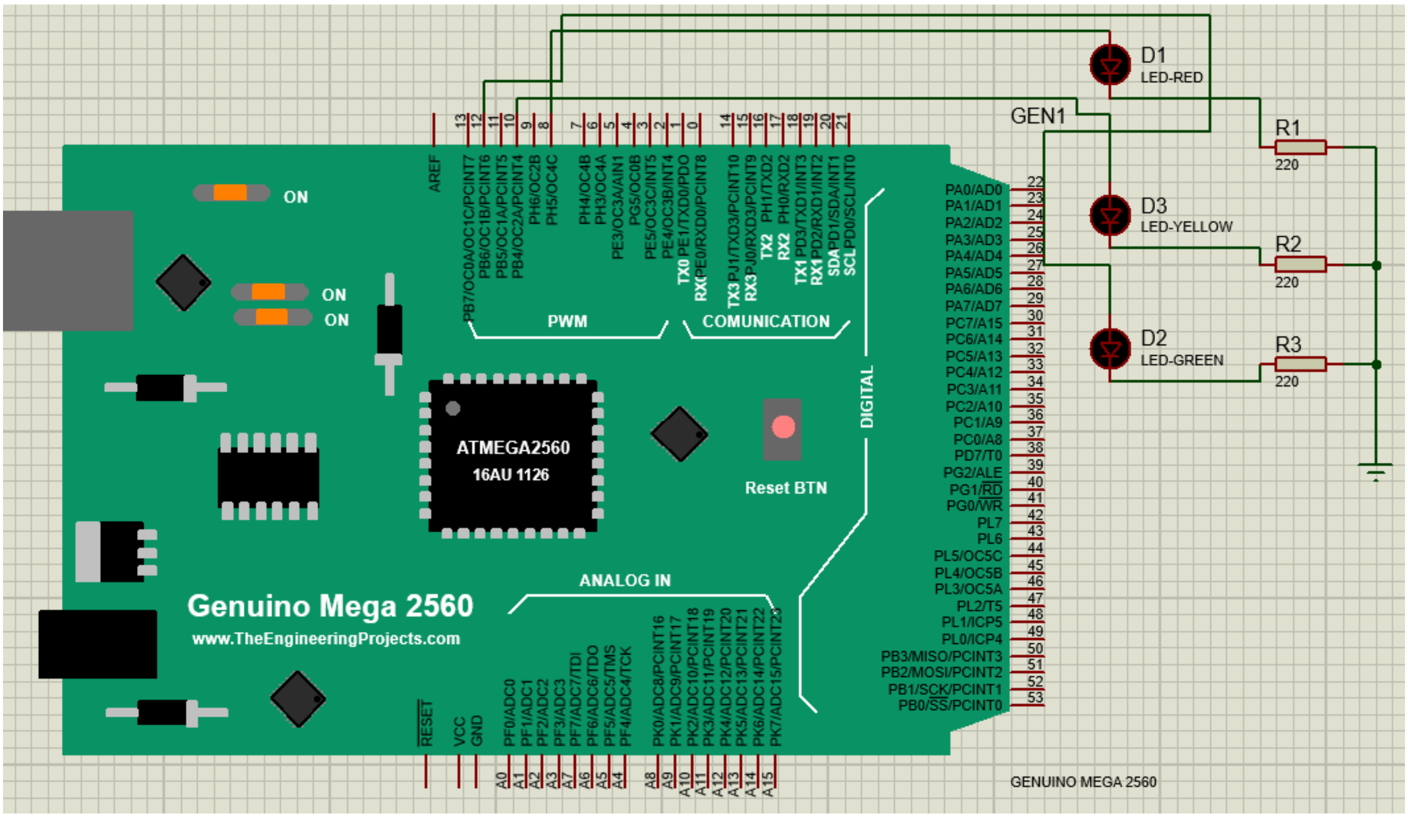


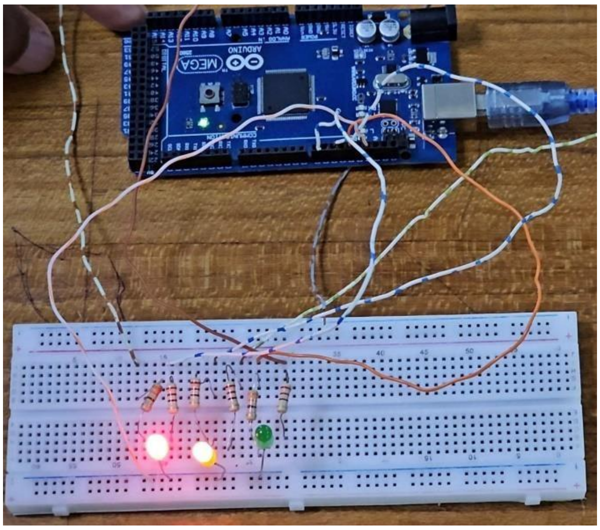
Figure 3: Simulation of a TLC using Arduino Mega

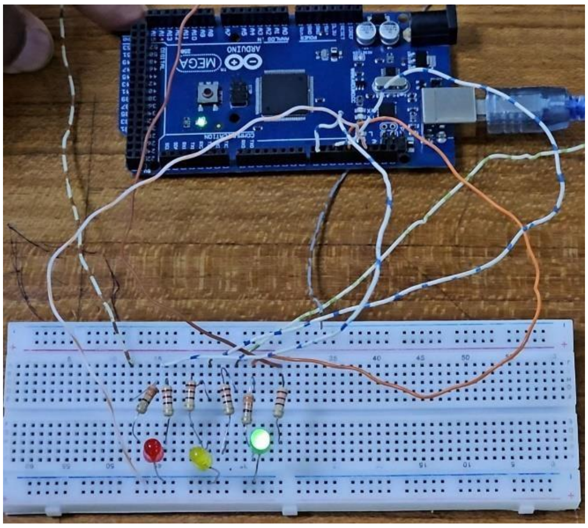
Code Implementation:

```
#define RED_PIN 8  //define name of pins used
#define YELLOW_PIN 10
#define GREEN_PIN 12

//define the delays for each traffic light color
int red_on = 3000; //3s delay
int red_yellow_on = 1000; //1s delay
int green_on = 3000; //3s delay
int green_blink = 500; //.5s delay
int yellow_on = 1000; //1s delay


int delay_timer (int milliseconds)
{
  int count = 0;
  while(1)
  {
    if(TCNT0 >= 16)  // Checking if 1 millisecond has passed
      {
        TCNT0=0;
        count++;
        if (count == milliseconds) //checking if required milliseconds delay has passed
```

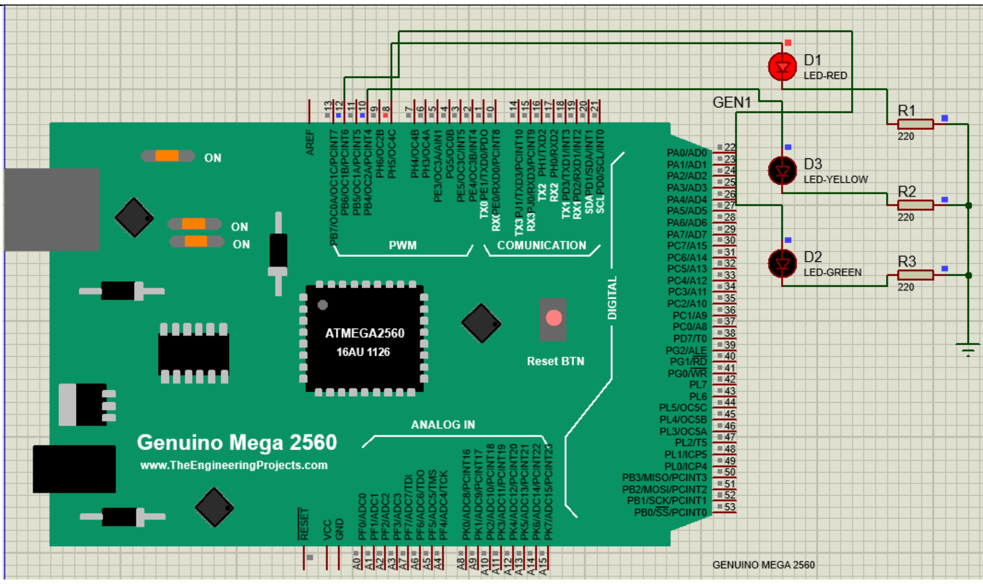
```
        {
            count=0;
            break; // exits the loop
        }
      }
  }
  return 0;
}

void setup() {
  // put your setup code here, to run once:
  pinMode(RED_PIN, OUTPUT);
  pinMode(YELLOW_PIN, OUTPUT);
  pinMode(GREEN_PIN, OUTPUT);

  TCCR0A = 0b00000000;
  TCCR0B = 0b00000101; //setting pre-scaler for timer clock
  TCNT0 = 0;
}

void loop() {
  // put your main code here, to run repeatedly:
  //to make red LED on
    digitalWrite(RED_PIN, HIGH);
    delay_timer(red_on);

    //to turn yellow LED on
    digitalWrite(YELLOW_PIN, HIGH);
    delay_timer(red_yellow_on);

    //turning off RED_PIN and YELLOW_PIN, and turning on greenLED
    digitalWrite(RED_PIN, LOW);
    digitalWrite(YELLOW_PIN, LOW);
    digitalWrite(GREEN_PIN, HIGH);
    delay_timer(green_on);
    digitalWrite(GREEN_PIN, LOW);

    //for turning green Led on and off for 3 times
    for(int i = 0; i < 3; i = i+1)
    {
    delay_timer(green_blink);
    digitalWrite(GREEN_PIN, HIGH);
    delay_timer(green_blink);
    digitalWrite(GREEN_PIN, LOW);
    }

    //for turning on yellow LED
    digitalWrite(YELLOW_PIN, HIGH);
    delay_timer(yellow_on);
    digitalWrite(YELLOW_PIN, LOW);
}
```

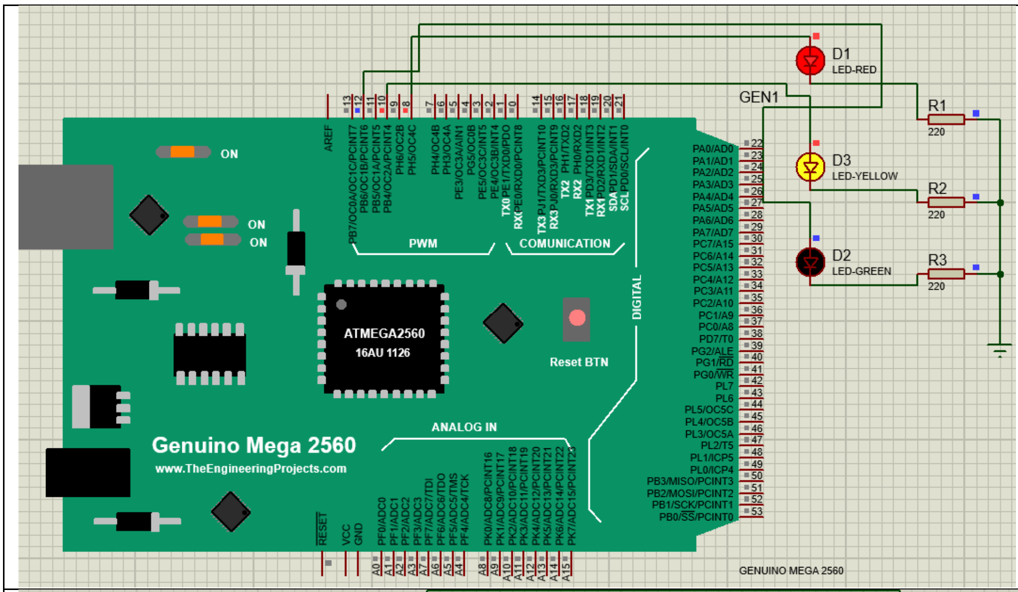
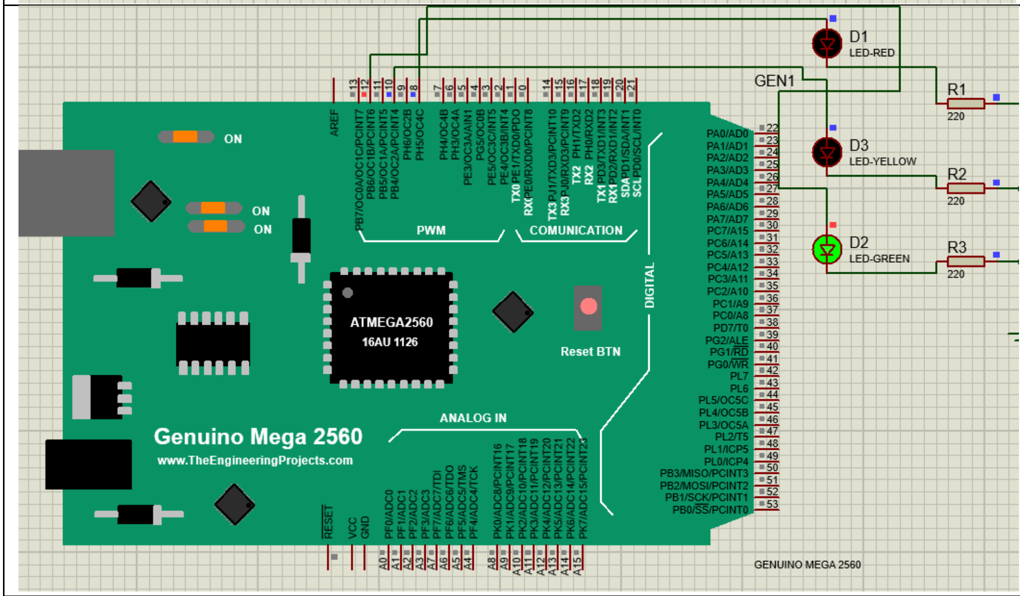
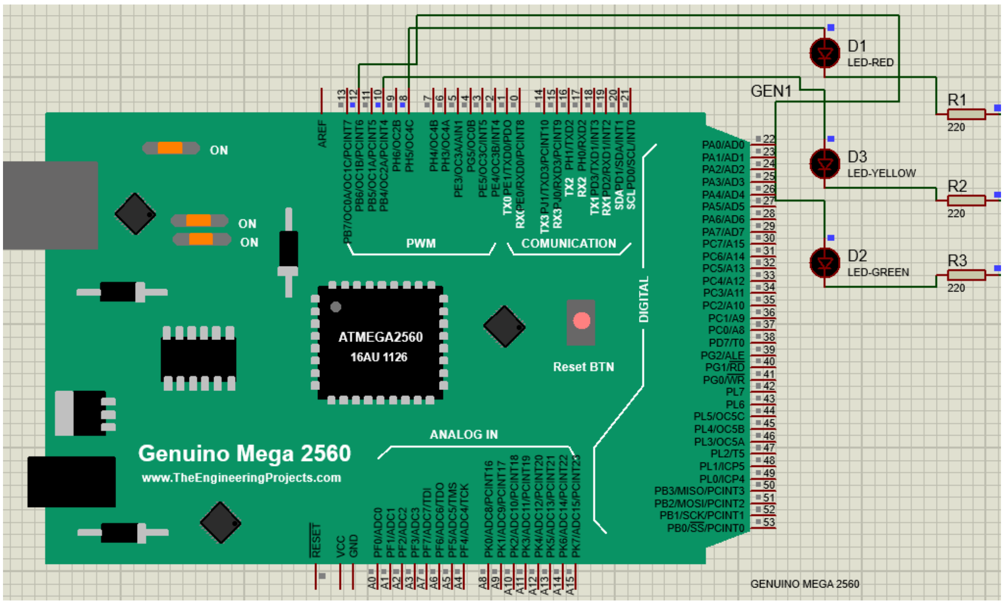
Hardware Outputs:

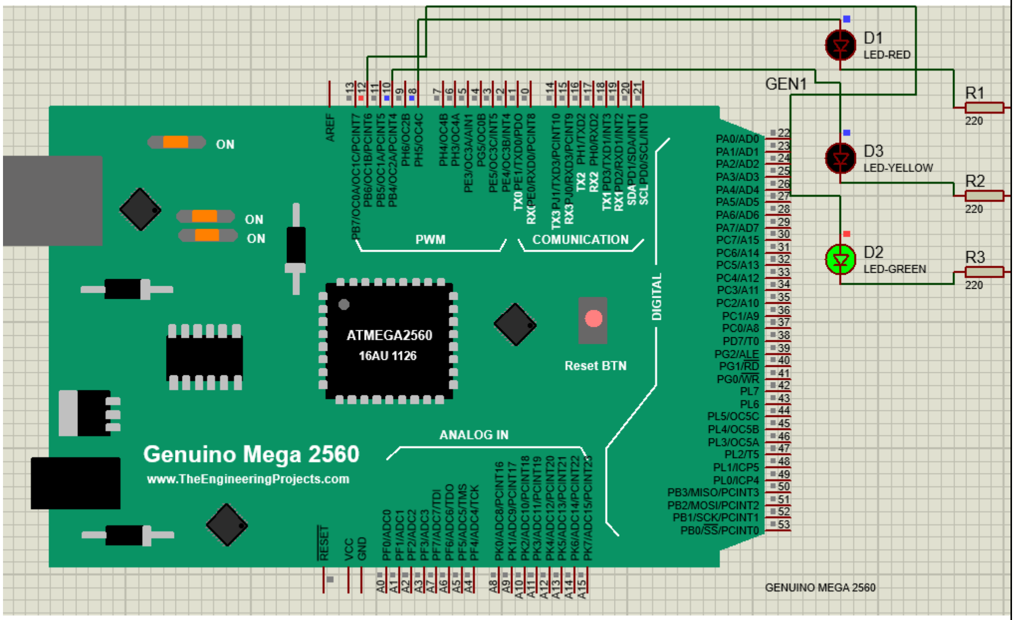
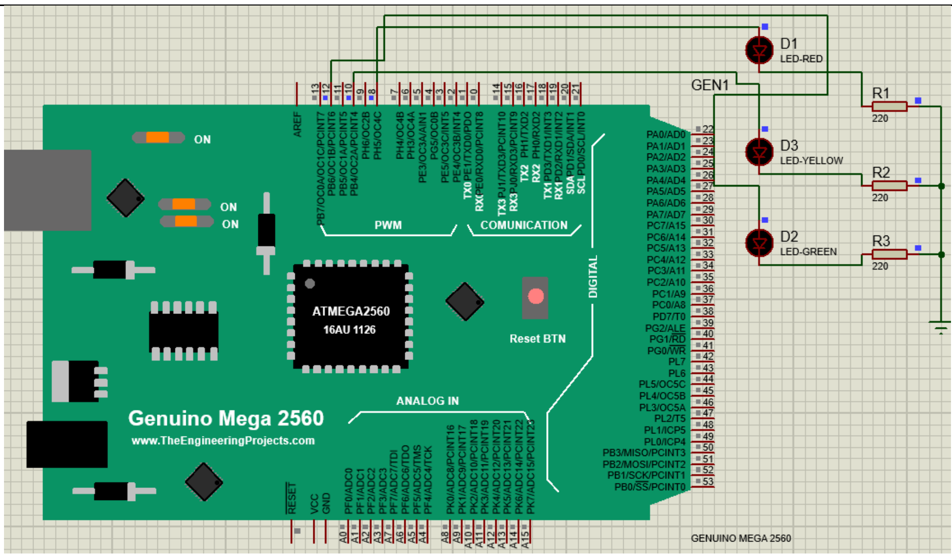
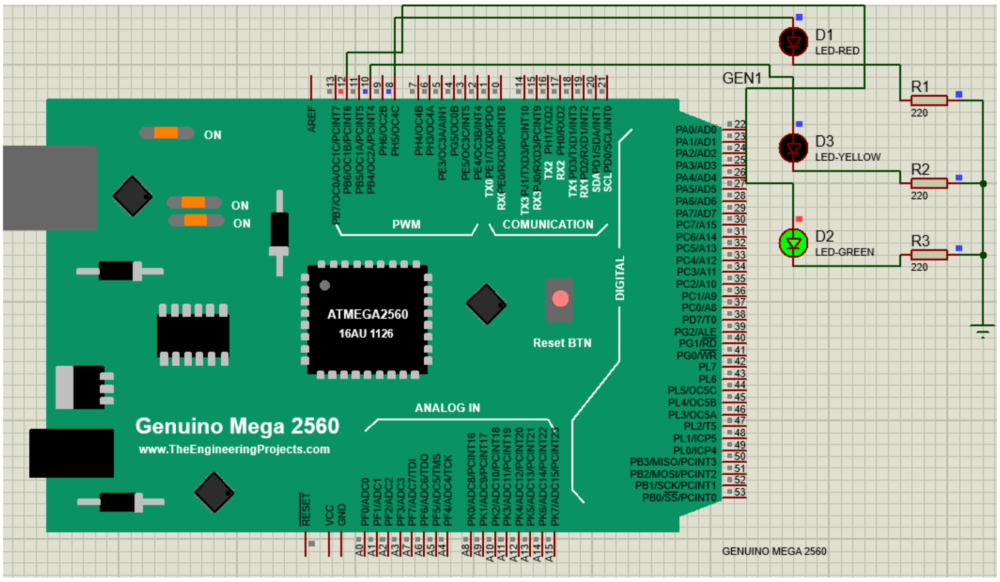


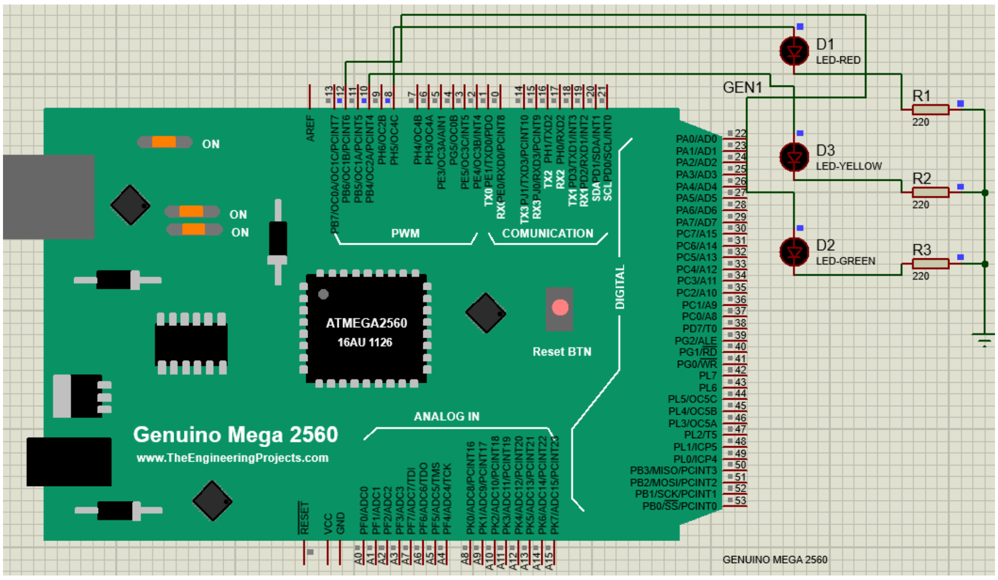
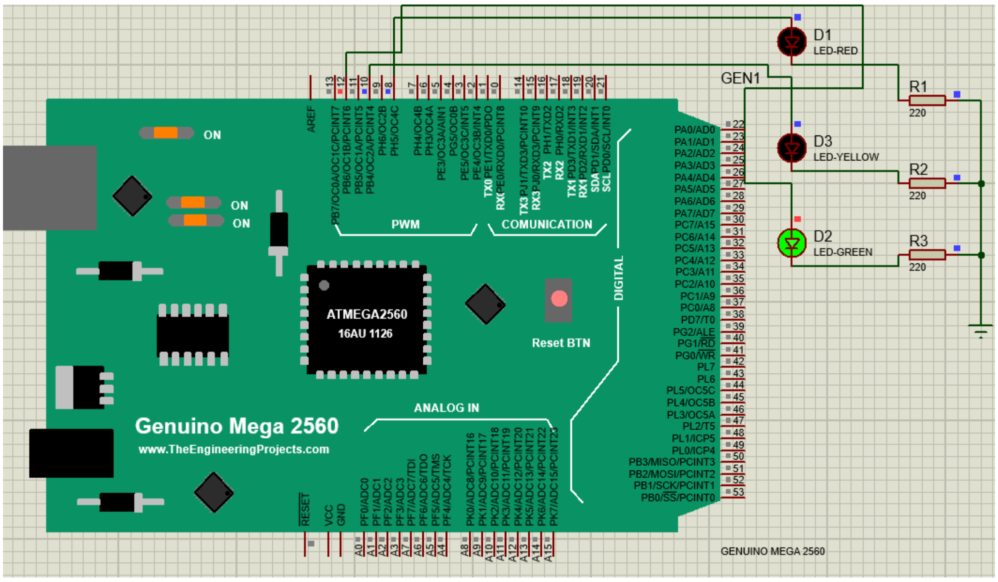
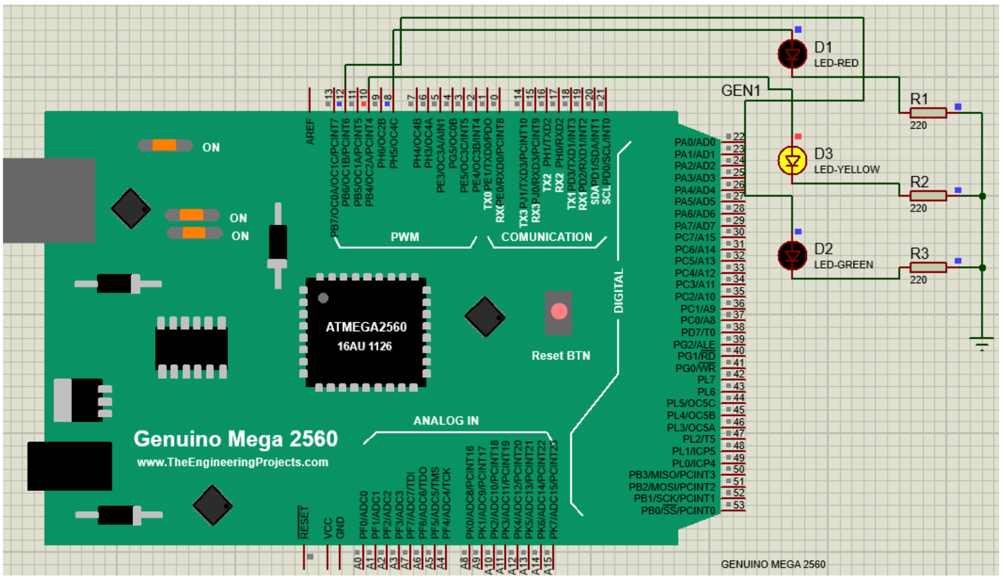


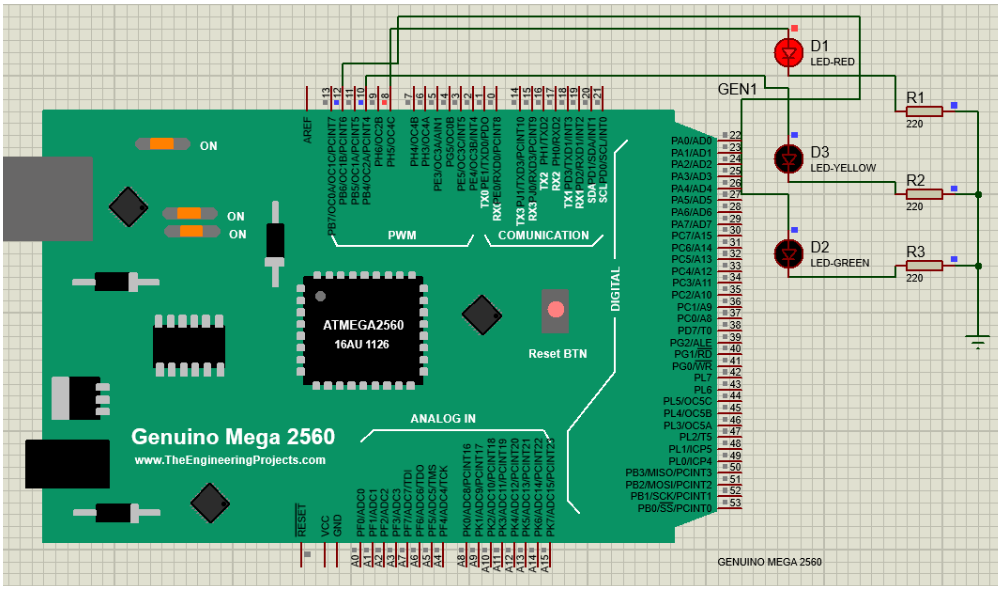
Simulation Outputs:

<table>
<tr><th>Output</th><th>Comment</th></tr>
<tr><td>
</td><td>Cycle: 1
Duration: 3s
RED LED: ON
YELLOW LED: OFF
GREEEN LED: OFF</td></tr>
</table>

	Cycle: 1 Duration: 1s RED LED: ON YELLOW LED: ON GREEEN LED: OFF
	Cycle: 1 Duration: 3s RED LED: OFF YELLOW LED: OFF GREEEN LED: ON
	Cycle: 1 Duration: 0.5s RED LED: OFF YELLOW LED: OFF GREEEN LED: OFF

	Cycle: 1 Duration: 0.5s RED LED: OFF YELLOW LED: OFF GREEEN LED: ON (Blink 1)
	Cycle: 1 Duration: 0.5s RED LED: OFF YELLOW LED: OFF GREEEN LED: OFF
	Cycle: 1 Duration: 0.5s RED LED: OFF YELLOW LED: OFF GREEEN LED: ON (Blink 2)

	Cycle: 1 Duration: 0.5s RED LED: OFF YELLOW LED: OFF GREEEN LED: OFF
	Cycle: 1 Duration: 0.5s RED LED: OFF YELLOW LED: OFF GREEEN LED: ON (Blink 3)
	Cycle: 1 Duration: 1s RED LED: OFF YELLOW LED: ON GREEEN LED: OFF

	Cycle: 2 Duration: 3s RED LED: ON YELLOW LED: OFF GREEEN LED: OFF (Repeat of cycle: 1)

Discussion:

The main objectives of this experiment were to get familiarized with Timers of Arduino Mega and use them for the implementation of a Traffic Light Control System which was accomplished properly. This experiment not only reinforced the understanding of microcontroller programming but also highlighted the critical role of timers in the precise control of automated systems like TCL.

In summary:

- The hardware implementation of a functional Traffic Light Control System utilizing timers on Arduino Mega was successfully done in the lab to manage the timing sequences of the traffic lights.
- The timer interrupts were configured, timer parameters were set properly, and integration of these elements was done.
- Simulation was successfully done using Proteus software according to the lab manual.
- Practical experience was gained in microcontroller programming while programming the Traffic Light Controller System using Arduino IDE.
- The results were verified by resembling the hardware implementation results and the simulation results.

Overall, this experiment has been successful in achieving its objectives as it helped the group members not only become familiarized with the use of timers in microcontrollers but also implement this knowledge in a real-world context. The skills and understanding gained from this experiment lay a strong foundation for future endeavors in microcontroller and embedded systems.

Conclusion: It can be said that the experiment was successfully done as all the results from the hardware implementation perfectly resemble the outputs of the simulations.

References:

[1] Arduino, “Arduino,” *Arduino.cc*, 2018. https://www.arduino.cc/ (Accessed: 15 June 2024).

[2] “[TUT] [C] Newbie’s Guide to AVR Timers,” *AVR Freaks*, May 20, 2007. https://www.avrfreaks.net/forum/tut-c-newbies-guide-avr-timers (Accessed: 15 June 2024).

[3] Mayank, “AVR Timers - TIMER0» maxEmbedded,” *maxEmbedded*, Jun. 24, 2011. https://maxembedded.com/2011/06/avr-timers-timer0/ (Accessed: 15 June 2024).

[4] Dermawan, M. and Meliala, S. (no date) *Design traffic light of HCSR04 sensor fuzzy logic method based on Arduino Mega 2560*, *International Journal of Engineering, Science and Information Technology*. Available at: https://www.ijesty.org/index.php/ijesty/article/view/386 (Accessed: 15 June 2024).

[5] Ismail, M.Z., Lutfi, A.Z.A. and Roslan, M.A.M. (2019) *Smart traffic light for emergency vehicle by using arduiono*, *AIP Publishing*. Available at: https://pubs.aip.org/aip/acp/article-abstract/2129/1/020099/1007182/Smart-traffic-light-for-emergency-vehicle-by-using (Accessed: 15 June 2024).